

## Difference Between JPA, Hibernate, and Spring Data JPA

| Feature                | JPA                             | Hibernate      | Spring Data JPA            |
|------------------------|---------------------------------|----------------|----------------------------|
| Type                   | Specification                   | ORM Framework  | Spring Module              |
| Developed By           | Oracle/Java Community (JSR 338) | Hibernate Team | Spring                     |
| Implementation         | No                              | Yes            | No                         |
| Purpose                | Defines persistence rules       | Implements JPA | Simplifies JPA usage       |
| SQL Generation         | No                              | Yes            | Uses Hibernate internally  |
| Boilerplate Code       | More                            | Moderate       | Very Less                  |
| Transaction Management | No                              | Manual         | Automatic (@Transactional) |
| CRUD Methods           | No                              | Session API    | JpaRepository              |

### Relationship

Java Application

|



Spring Data JPA

|



Hibernate

|



JPA Specification

|



## MySQL Database

### 1. Java Persistence API (JPA)

#### Definition

JPA (**Java Persistence API**) is a **specification** that defines how Java objects should be stored and retrieved from relational databases.

It only defines **interfaces and annotations**.

**It does not contain any implementation.**

Examples:

@Entity

@Table

@Id

@Column

### 2. Hibernate

#### Definition

Hibernate is an **ORM (Object Relational Mapping) Framework**.

It is the **most popular implementation of JPA**.

Hibernate actually performs the database operations.

Example:

```
Session session = factory.openSession();
```

```
Transaction tx = session.beginTransaction();
```

```
session.save(employee);
```

```
tx.commit();
```

```
session.close();
```

Hibernate converts Java objects into SQL queries automatically.

### 3. Spring Data JPA

#### Definition

Spring Data JPA is a Spring framework module built on top of JPA.

**It does not implement JPA.**

Instead, it reduces boilerplate code.

Example:

Instead of writing

```
Session session = factory.openSession();
```

```
session.save(employee);
```

simply write:

```
employeeRepository.save(employee);
```